



MINISTRY OF EDUCATION
STATE DEPARTMENT OF EARLY LEARNING AND BASIC EDUCATION

Curriculum Design Interpretation





Components of a Curriculum Design

Activity:

Brainstorm on the components of a curriculum
design



Components of a Curriculum Design

National Goals of Education

Level Learning Outcomes

Subject /General Learning Outcomes

Essence Statement

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Question(s)

Core competencies to be developed

PCIs	Values
Links to other subjects	Suggested Community Service Learning activities

Assessment Rubrics

Appendix:

Suggested assessment methods	Suggested learning resources	Suggested Non-formal activities
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National Goals of Education

1. Foster nationalism, patriotism and promote national unity
2. Promote social, economic, technological and industrial needs for national development
3. Promote individual development and self-fulfilment
4. Promote sound moral and religious values
5. Promote social equity and responsibility
6. Promote respect for and development of Kenya's rich and varied cultures
7. Promote international consciousness and foster positive attitudes towards other nations
8. Promote positive attitudes towards good health and environmental protection.



ACHIEVEMENT OF NATIONAL GOALS OF EDUCATION THROUGH THE COMPETENCY BASED CURRICULUM

- ❑ The Competency Based Curriculum focuses on acquisition of the desired **knowledge, skills, values and attitudes** to cope with contemporary life challenges.
- ❑ **Learning outcomes** are derived from the National Goals of Education.
- ❑ **The achievement of learning outcomes** leads to desired behavioral change.



LEARNING OUTCOMES

- ❑ Learning outcomes are the **end result** or what the learner is able to demonstrate upon completion of a lesson, sub strand, strand or level.
- ❑ Learning outcomes are expressed through the application of **knowledge, skills and attitudes**.
- ❑ Learning outcomes express what a learner is expected **to do** after going through a learning experience.
- ❑ They indicate the expected **observable behavioural changes** in the learner after a learning experience.



Categories of Learning Outcomes in the Curriculum Design

Activity: Studying Categories of Learning Outcomes in Curriculum Design

1. Use a sample curriculum design (any learning area)
2. Identify and Study the following categories of learning outcomes:
 - A: Level learning outcomes**
 - B: Subject /General learning outcomes**
 - C: Specific learning outcomes**
3. What is the difference between the three levels of learning outcomes.



LEVEL LEARNING OUTCOMES FOR MIDDLE SCHOOL EDUCATION

By the end of Middle School, the learner should be able to:

- 1) apply literacy, numeracy skills and logical thinking appropriately in self-expression,
- 2) communicate effectively in diverse contexts,
- 3) apply digital literacy skills appropriately for communication and learning in day-to-day life,
- 4) practise hygiene, appropriate sanitation and nutrition to promote health,
- 5) explore, manipulate, manage and conserve the environment effectively for learning and sustainable development,
- 6) demonstrate ethical behaviour and exhibit good citizenship as a civic responsibility,
- 7) demonstrate social skills, spiritual and moral values for peaceful co-existence,
- 8) demonstrate appreciation of the country's rich, diverse cultural heritage for harmonious co-existence,
- 9) manage Pertinent and Contemporary Issues in society effectively



Activity: Relationship between the learning outcomes and the National Goals of Education

1. Pick on any National Goal of Education
2. Identify one level learning outcome for middle school that will help realise the National Goal of Education in N0 1 above.
3. Identify one subject/general learning outcome that will aid realization of the level learning outcome in N0. 2 above.
4. From any learning area, pick one specific learning outcome that will help achieve the subject general learning outcome identified in 3 above
5. Wrap up the session by discussing the relationship that exists between the national goals and various outcomes. Participants should also not that all outcomes are derived from the national goals.

Progression of Learning outcomes

National Goals of Education



Level learning outcomes



General learning outcomes



Specific learning outcomes

Essence statements

Statements that describe:

- ☐ Why a subject is taught at a certain level
- ☐ The strands in the subject
- ☐ Learning approaches to be used
- ☐ Theoretical underpinning for the subject



Specific learning outcomes

Since CBC is **learner centred and activity oriented**, learning outcomes are designed following the same trend.

The outcome statement includes **a verb** that will clearly illustrate what **action** learners need to engage in to demonstrate their competencies.

The **action** could demonstrate a change at the level of knowledge, a skill or an attitude.



Example:

By the end of the sub strand the learner should be able to:

- a) **identify** types of soil erosion in the environment (**knowledge**)
- b) **control** soil erosion in the school environment (**skills**)
- c) **appreciate** the role of soil conservation in maintaining agricultural environment (**attitude**).

The verbs represent the three domains of learning.

A good learning outcome begins with an **action verb**, followed by the **object of the verb** and a phrase that gives the **context**.

Other verbs that can be use instead of appreciate include acknowledge, advocate, value, embrace, recognise among others



Characteristics of good specific learning outcomes

- a) The specific learning outcomes should be observable.
- b) Specific learning outcomes are agreeable to assessment.
- c) Avoids vague immeasurable terms like know, understand, learn etc

Activity

Studying Specific Learning Outcomes in Curriculum Design

1. Use a sample curriculum design (any learning area)
2. Identify and study specific learning outcomes from several subjects
3. Discuss the verbs used in creating them

Writing Learning outcomes

Begin each specific learning outcome with an **action verb**, followed by the **object of the verb** followed by a **phrase that gives the context**.

Preferably use only one verb per learning outcome.

Avoid vague terms like **know, understand, learn, be familiar with, be exposed to, be acquainted with, and be aware of, demonstrate understanding**.

The specific learning outcomes must be observable.

Ensure that the specific learning outcomes are capable of being assessed

A specific Learning Outcome must develop the following:

KNOWLEDGE

- Acquisition of facts, ideas, concepts; their application, synthesis to get new ideas and evaluation of the information

SKILLS

- Using the knowledge acquired, a learner is able to apply it to perform certain tasks against a given standard. The standard to which a learner should perform the task is referred to as a competence.

VALUES

- Desirable beliefs and personal attributes that a learner acquires through knowledge and skill that influence performance.

A Specific Learning Outcome must develop the following:



Bloom's Level	Key Verbs (keywords)	Example Learning outcome
Create	design, formulate, build, invent, create, compose, generate, derive, modify, develop.	Design a garment for a new born baby
Evaluate	choose, support, relate, determine, defend, judge, grade, compare, contrast, argue, justify, support, convince, select, evaluate.	Choose foods rich in vitamin C in their locality
Analyze	classify, break down, categorize, analyze, diagram, illustrate, criticize, simplify, associate.	differentiate between macro and micro nutrients found in foods.
Apply	calculate, predict, apply, solve, illustrate, use, demonstrate, determine, model, perform, present	calculate BMI for promoting health.



Activity: Development of Specific Learning Outcomes

Work in pairs:

Choose a sub strand in your learning area (Grade 6 level)

Develop specific learning outcomes to achieve the concept within in a lesson.



Example of Verbs used in the development of SLO's Knowledge, Skills and Attitudes

By the end of the sub strand the learner should be able to :

identify types of waste in the school environment

collect waste in the school to care for the environment

appreciate cleanliness in the school environment

The verbs represent the three domains of learning including,

knowledge, skills and attitudes.



The object in specific learning outcomes

The outcome also has an object, which answers the question
“what”

identify **types of waste** in the school environment
collect **waste** in the school to care for the environment
appreciate **cleanliness** in the school environment



The Context in learning outcomes

The outcome also has a context, which answers the question “**why**”, “**where**”, “**how**” and “**when**”

identify types of waste **in the school environment.** (*where*)

collect waste in the school **to care for the environment.** (*why*)

appreciate cleanliness **in the school environment.** (*where*)



Activity

1. Choose a sub strand in a learning area of your choice.
2. Develop specific learning outcomes
3. Present them to the plenary



Learning Experiences

Activity: Group Discussion

1. Identify the formal, non-formal and informal experiences/interactions that can enable students to acquire the desired competencies.
2. Write them on a manila paper.
3. Post them on walls of the training room.
4. Participants to take a gallery walk to view what others have written.



Learning Experiences

Activity

Can you remember any **learning experience** during your school or college days?

What makes it memorable?

How did it make you feel?

Buzz with your neighbor and share in the plenary.

Types of Learning Experiences

Formal

- Organized within classroom or other school contexts (e.g. the field, school garden etc.)
- The learning experiences are **dictated by specific learning outcomes in the curriculum design.**

Non-formal

- Critical for the learner to continue applying what she/he has learnt in scenarios outside the classroom.
- This include clubs and societies. The learner is able to apply **knowledge/skills/values and attitudes** they have learnt formally outside the class.

Informal

- Entails **general social interactions** at school, home and in the wider community.
- The community plays a key role in shaping the moral nature/compass of the learner
- These experiences offer the learner a forum to **model values.**

Suggested Learning Experiences

Learning experiences enable the learner to:

- ❖ acquire the intended core competencies
- ❖ engage deeply in the learnt concepts
- ❖ reflect on the learning process (self assessment)
- ❖ interact with others during the learning process.

Factors to consider in selection of learning experiences





Meaning ?

Learning experiences are **activities or tasks** that the trainee should carry out in order to develop desired **knowledge, skills, values** and **attitudes**

Learning experiences focus on developing the trainees **cognitive, physical and emotional domains**.

They are interactions in which learning takes place.

“Interaction between the learner and external conditions in the environment to which he/she can react.” Ralph Tyler

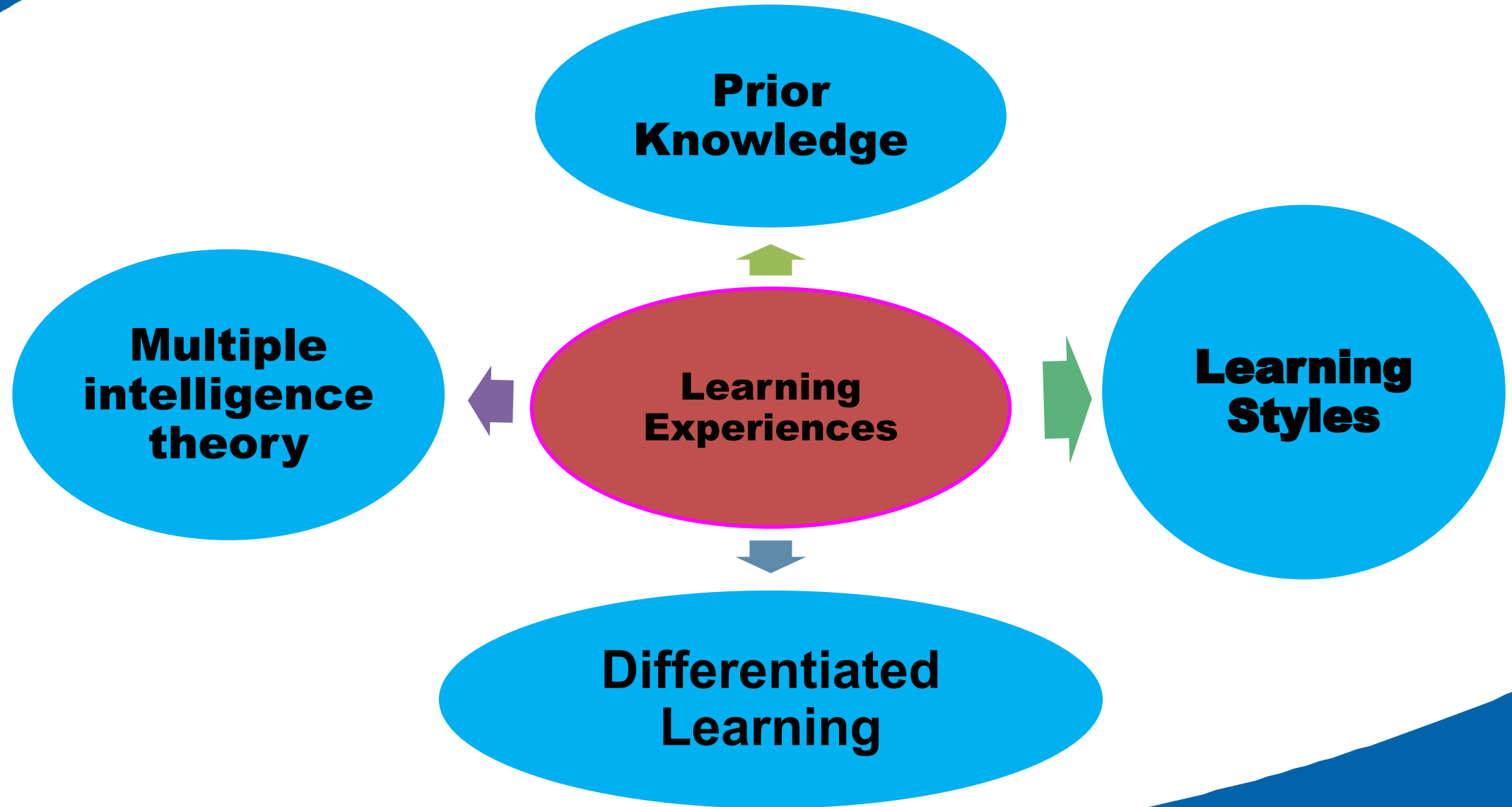


Purpose of Learning Experiences

They enable the learner to:

- ❖ acquire and apply knowledge and develop skills, values and attitudes
- ❖ acquire and apply the intended core competences
- ❖ model the core values
- ❖ self-regulate and evaluate their learning
- ❖ engage deeply in the learnt concepts
- ❖ reflect on the learning process
- ❖ interact with others during the learning process

What informs the learning Experiences?







Key Inquiry Questions (KIQ)

WHAT ARE THEY FOR ?

- ❖ Asked to stimulate ongoing thinking and inquiry
- ❖ Raise more questions
- ❖ Spark discussion and debate
- ❖ Demand justification and support
- ❖ Responses may change as understanding deepens

ACTIVITY:WHAT IS A BADLY STATED KEY INQUIRY QUESTION ?

Participants to brainstorm and give examples of the following:

1. A question that can be answered with a yes or no or close ended question such as “Do you go to school?”
2. A question that can be answered without support.
3. A blurry (undefined/clearly stated)



CREATING KEY INQUIRY QUESTIONS

- First, consider the focus of your specific learning outcome as addressed in the curriculum design.
- Then, examine the **theme or concept** in the curriculum that must be addressed and brainstorm on questions that would enable learners to think about the concept without dictating the direction or outcome of their thinking

Utilize the six typical question words: Who? What? Where? When? How?

Who? & why? Questions are referred to as interrogative and procedural questions that illicit deep thinking and are most recommended for use.

TYPES OF KIQ

Questions that Hook

- Asked to interest learners around a new concept
- May spark curiosity, questions, or debate

Questions that Lead

- Have a "correct" answer
- Support recall and information finding
- Asked until *the* answer is given
- Require no (or minimal) support

Questions that Guide

- Asked to encourage and guide exploration of a concept
- Point towards desired knowledge and skill (but not necessarily to a single answer)
- Generally require some explanation and support



Activity

In groups, pick a sub strand from any subject of choice and analyse the KIQs
Develop a few of your own that would elicit deep thinking in the learner during a lesson.



Core competences to be developed

1. Communication and Collaboration
2. Self-efficacy
3. Critical Thinking and Problem Solving
4. Creativity and Imagination
5. Citizenship
6. Digital Literacy
7. Learning to Learn

NB: The seven core competencies are to be learnt in all the learning areas.
The **indicators** for the specific core competence that shall be developed in the learning experiences. (Refer to CBA indicators for core competences



Activity: Core competencies

1. Brainstorm on ways of ensuring core competencies are integrated in the various sub strands during the learning process in your subject.
2. Wrap up the activity by going through the points raised and the next slide.

How to integrate the core competencies



This is done during **lesson planning**. The teacher should consider ;

The core
competence to
be developed

- The *indicators* for the specific core competence that shall be developed in the learning experiences
- The **specific learning outcomes** of the lesson- do they accommodate the core competence to be developed?
- **Learning experiences** that will allow the learner to develop the core competences
- **Learning resources** that will help the learner to develop the core competence –eg use of ICTs to acquire digital literacy
- The **class organization** that will help the learner develop the core competence





Values

Values are modelled not taught

“Children adopt the values they experience, not the values that are dictated to them” Dr. Neil Hawkes

Brainstorm some indicators of the core values below:

1. Respect
2. Patriotism
3. Social Justice
4. Unity
5. Love
6. Responsibility
7. Peace
8. Integrity

(refer to CBA Indicators of core values)



Activity: Values

Participants to brainstorm on ways through which values can be inculcated, through the formal, non formal and informal curriculum.

How to develop values in the learner

Formally

Selection of learning experiences where the learner will be guided to model the targeted value
e.g. during group work, debates, dramatisation, talks, sharing of resources, concepts like etiquette

Non -formally

Guiding learner to model values within club activities – the teacher will grasp opportunities to guide the learner to model expected social norms when interacting with peers. This can also be achieved during assembly, guest speakers

Informally

The class teacher and school administration, and school community should create 'a culture' of expected social norms that everyone should model. Informal interactions can be through talking walls and trees, modelling proper dressing, polite language, sharing resources among others



Link to other subjects

No subject stands alone in the curriculum. Concepts run across subjects and it is important that the teacher makes reference to **knowledge** or **skills** learnt in another subject.



Assessment rubrics

Every sub strand should have an assessment rubric

This is a **criterion referenced tool** that assists the teacher to assess **the process** or the **completed work** done by a learner

The rubric has **4 measurement scales** with descriptors which describe the **performance levels**



Example: Social studies

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Question (S)
1.0 NATURAL AND THE BUILT ENVIRONMENTS IN AFRICA	1.3 The Earth and the Solar System (6 Lessons)	By the end of the sub strand, the learner should be able to: a) Describe the origin, size, shape and position of the earth in the solar system b) Examine the effects of rotation and revolution of the earth and its effects on human activities c) Illustrate the internal structure of the earth in the solar system d) Appreciate the effects of rotation and revolution of the earth on human activities	Learners are guided to: - Think, pair and share on the origin of the earth - Share stories on the origin of the earth from their communities - Brainstorm in groups on the origin, size, shape of the earth and share in class - Use relevant digital resources to find out about the theories explaining the origin of the earth (the passing star theory and the Nebula theory) - Do library research on the position of the earth in the solar system and write short notes - View a video clip on the earth and the solar system and discuss with peers - Draw the solar system and indicate the position of the earth as the home of humankind - Model the solar system and show the position of the earth and display in class - Use locally available materials to model and demonstrate the effects of rotation and revolution of the earth and its effects on human activities - Discuss in groups the internal structure of the earth and do a class presentation(inner core, mantle and crust)	How did the earth come into being?

ASSESSMENT RUBRICS

Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to describe the origin, size, shape and position of the earth in the solar system	Accurately and proficiently describes the origin, size, shape and position of the earth in the solar system	Accurately describes the origin, size, shape and position of the earth in the solar system	Partially describes the origin, size, shape and position of the earth in the solar system	Has challenges describing the origin, size, shape and position of the earth in the solar system
Ability to examine the effects of rotation and revolution of the earth and its effects on human activities	Correctly and comprehensively examines the effects of rotation and revolution of the earth and its effects on human activities	Correctly examines the effects of rotation and revolution of the earth and its effects on human activities	Correctly examines some of the effects of rotation and revolution of the earth and its effects on human activities	Has difficulties in examining the effects of rotation and revolution of the earth and its effects on human activities
Ability to illustrate the internal structure of the earth in the solar system	Correctly and creatively illustrates the internal structure of the earth in the solar system	Correctly illustrates the internal structure of the earth in the solar system	Correctly illustrates some aspects of the internal structure of the earth in the solar system	Has challenges illustrating the internal structure of the earth in the solar system



THANK YOU !